EXAM-ORIENTED STUDY NOTES

Basic Concept of Networking and Data Communications

LAN · OSI 7 Layers · Topologies · Protocols · Ethernet · ARCNET · TCP/IP

Coverage:

1. Introduction to Networking & Data Communication
2. Local Area Network (LAN)
3. OSI Reference Model (7 Layers)
4. Network Topologies
5. Network Protocols
6. Ethernet
7. ARCNET
8. TCP/IP Model
9. Quick Revision Section

Prepared for Competitive Examinations · University Examinations · Teacher Recruitment Examinations

Chapter 1: Introduction to Networking and Data Communication

1.1 Definition of Computer Network

A **computer network** is an interconnected collection of autonomous computers and devices linked together by communication channels (wired or wireless) to share data, resources (such as printers and files), and services with one another.

> **Key Definition**
> A network = two or more devices + communication links + protocols that allow sharing of information and resources.

1.2 Data Communication Basics

Data Communication refers to the exchange of data between two devices through a transmission medium such as a wire cable. For data communication to occur, the communicating devices must be part of a communication system made up of hardware (physical equipment) and software (programs).

Key Terms

Term	Meaning
Data	Raw facts/figures represented in a form that can be processed by a computer (text, numbers, images, audio, video).
Signal	Electrical or electromagnetic representation of data used during transmission (analog or digital).
Transmission	The act of sending data/signal from sender to receiver through a medium.
Bandwidth	The data carrying capacity of a channel, measured in bits per second (bps).

1.3 Components of a Data Communication System

Every data communication system has **five basic components**:

1. **Message** – the information to be communicated (text, number, picture, audio, video).
2. **Sender** – the device that sends the message (computer, workstation, telephone, etc.).
3. **Receiver** – the device that receives the message.
4. **Medium / Transmission Medium** – the physical path through which the message travels (wired or wireless).
5. **Protocol** – the set of rules that governs data communication and represents an agreement between the communicating devices.



Fig. 1.1 — Basic Data Communication System

1.4 Characteristics of Effective Data Communication

Characteristic	Description
Delivery	Data must be delivered to the correct destination only.
Accuracy	Data must be delivered accurately, without alteration/errors.
Timeliness	Data must be delivered in a timely manner; late delivery is useless (e.g., real-time audio/video).
Jitter	Uneven delay in delivery of packets; should be minimal, especially for audio/video data.

1.5 Need and Importance of Networking

- **Resource Sharing** – sharing hardware (printers, scanners) and software among users.
- **Data/File Sharing** – easy and fast transfer of files between users.
- **Communication** – enables email, messaging, video conferencing.
- **Centralized Data Management** – data can be stored and managed centrally on servers.
- **Cost Reduction** – reduces hardware/software duplication costs.
- **Reliability** – data can be backed up on multiple machines, increasing reliability.
- **Scalability** – networks can grow by adding new devices/users.

1.6 Communication Models

Two principal models describe how communication takes place between networked systems:

Model	Description
Simplex	Communication is one-directional only; one device transmits, the other only receives (e.g., Television broadcast, keyboard to monitor).
Half-Duplex	Both devices can transmit and receive, but not at the same time (e.g., Walkie-talkie).
Full-Duplex	Both devices can transmit and receive simultaneously (e.g., Telephone network).

Chapter Summary

Data communication is the exchange of data between devices via a transmission medium, governed by protocols. The five components are Message, Sender, Receiver, Medium, and Protocol. Communication can occur in simplex, half-duplex, or full-duplex mode. Networking is essential for resource sharing, communication, and centralized data management.

Chapter 2: Local Area Network (LAN)

2.1 Definition and Characteristics

A **Local Area Network (LAN)** is a network that connects computers and devices within a limited geographical area such as a single building, office, school, or campus.

Characteristic	Typical Value / Description
Geographical Coverage	Small area — up to a few kilometres (single building/campus)
Ownership	Privately owned by an organization
Data Transfer Rate	High speed (typically 10 Mbps to 10 Gbps)
Error Rate	Very low, as transmission distance is short
Transmission Medium	Twisted pair cable, coaxial cable, optical fibre, Wi-Fi
Cost	Relatively low setup and maintenance cost

2.2 Features of LAN

- Confined to a small geographical area.
- High speed of data transmission.
- Low error rates due to short distances.
- Allows sharing of hardware resources (printers, scanners) and software.
- Usually owned, controlled, and managed by a single organization.
- Can be wired (Ethernet) or wireless (Wi-Fi).

2.3 Advantages and Disadvantages of LAN

Advantages	Disadvantages
Resource sharing (printers, storage, internet)	Limited geographical coverage
High speed data transfer	Initial setup cost for cabling/devices
Centralized data storage and backup	Security risks if not properly managed
Easy communication among users (email, messaging)	Network failure can disrupt the whole system (depends on topology)
Centralized software management/updates	Requires skilled administrator for maintenance

2.4 Applications of LAN

- File and resource sharing in offices, schools, and colleges.
- Shared access to printers and scanners in an organization.
- Internal communication system (intranet, email, messaging).
- Centralized database access for businesses (e.g., banking, billing).
- Computer laboratories in educational institutions.

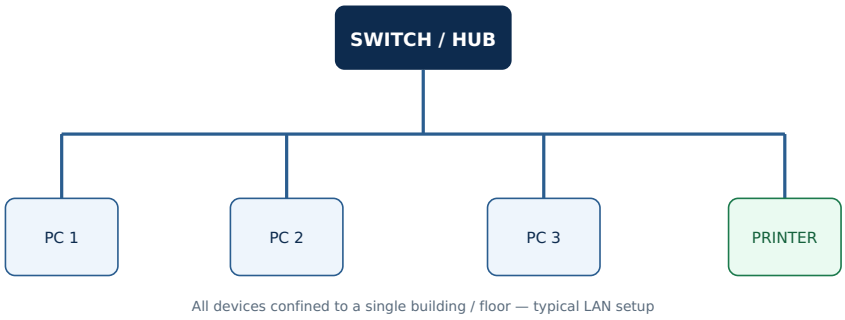


Fig. 2.1 — Typical LAN Structure

Chapter Summary

A LAN is a privately-owned, high-speed network confined to a small geographic area, used mainly for resource and file sharing within an organization.

Chapter 3: OSI Reference Model (7 Layers)

3.1 Introduction to OSI Model

The **Open Systems Interconnection (OSI)** model was developed by the **ISO (International Organization for Standardization)** in 1984. It is a conceptual framework that divides network communication into **seven layers**, each responsible for a specific function, enabling different systems to communicate using standard protocols.

The OSI model does not perform any function in the actual communication process by itself — it is a **reference model** that describes how data communication should logically take place. Real protocol suites (such as TCP/IP) are designed and understood with reference to this layered structure.

Attribute	Detail
Full Form	Open Systems Interconnection
Developed By	ISO (International Organization for Standardization)
Year of Publication	1984
Number of Layers	7
Nature	Theoretical / Conceptual reference model (not directly implemented as-is)
Layer Grouping	Upper layers (5,6,7) – Application/User support; Lower layers (1,2,3,4) – Data Transport support

3.2 Purpose and Importance

- Provides a standard for different vendors/manufacturers to build interoperable network devices and software.
- Breaks complex networking tasks into smaller, manageable layers.
- Helps in troubleshooting — problems can be isolated to a particular layer.
- Allows modular design — changes in one layer don't usually affect others.
- Serves as a teaching/reference model for understanding network operations.

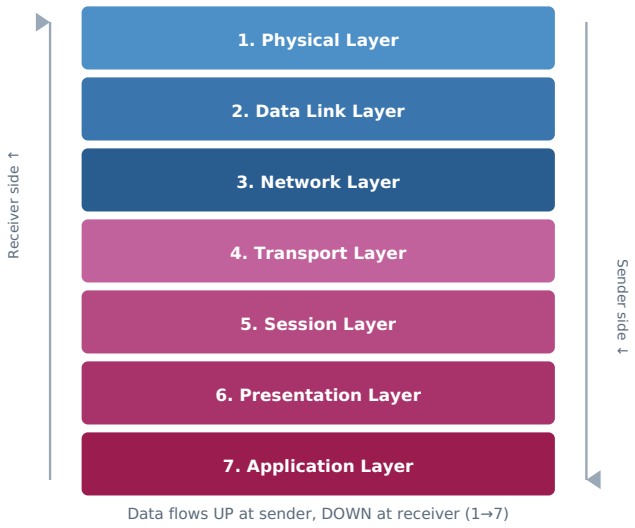


Fig. 3.1 — Complete OSI 7-Layer Model

3.3 Detailed Explanation of Each Layer

Layer 1: Physical Layer

- Deals with the transmission of raw **bits** (0s and 1s) over a physical transmission medium.
- Defines cable types, connectors, voltage levels, bit rate, and physical topology.
- Concerned with mechanical and electrical specifications of the network.

Function	Description
Bit Representation	Encodes bits into signals (e.g., voltage levels, light pulses, radio waves)
Data Rate / Bit Rate	Determines the number of bits transmitted per second
Synchronization of Bits	Sender and receiver clocks must be synchronized at the bit level
Line Configuration	Point-to-point or multipoint connection between devices
Physical Topology	How devices are physically arranged (bus, star, ring, mesh, etc.)
Transmission Mode	Simplex, half-duplex, or full-duplex

Devices/Protocols: Hubs, repeaters, cables (coaxial, twisted pair, fibre optic), connectors (RJ-45), NIC (Network Interface Card).

Unit of Data: Bit

Layer 2: Data Link Layer

- Responsible for node-to-node delivery of data, organizing bits into **frames**.
- Performs **error detection/correction** and **flow control**.
- Handles **MAC addressing** and media access control (e.g., CSMA/CD).
- Divided into two sub-layers: LLC (Logical Link Control) and MAC (Media Access Control).

Function	Description
Framing	Divides the stream of bits into manageable units called frames
Physical Addressing	Adds sender/receiver MAC address (hardware address) to the frame header
Flow Control	Prevents a fast sender from overwhelming a slow receiver
Error Control	Detects and retransmits damaged or lost frames (uses CRC/checksum)
Access Control	Determines which device has control over the shared link at a given time

Sub-layers of Data Link Layer

LLC (Logical Link Control): Handles flow control, error control, and identifies network layer protocols.
MAC (Media Access Control): Handles addressing and controls access to the shared physical medium.

What is a Trailer?

A **trailer** is the part of data added at the **end** of a Protocol Data Unit (mainly at the Data Link Layer), as opposed to a **header**, which is added at the beginning. It mainly carries error-checking information such as the **FCS (Frame Check Sequence)**, computed using CRC, so the receiver can verify whether the frame was received correctly.

Devices/Protocols: Switches, bridges, NIC; Ethernet, PPP, HDLC, Frame Relay.

Unit of Data: Frame

Layer 3: Network Layer

- Responsible for source-to-destination delivery of **packets** across multiple networks (routing).
- Uses **logical addressing (IP addresses)**.
- Performs path determination via routing algorithms.

Function	Description
Logical Addressing	Adds source and destination IP addresses to the packet
Routing	Selects the best path for the packet to travel from source to destination
Packetizing	Encapsulates data received from the transport layer into packets
Fragmentation	Breaks packets into smaller units if required by the underlying network

Devices/Protocols: Routers, Layer-3 switches; IP, ICMP, ARP, RIP, OSPF.

Unit of Data: Packet

Layer 4: Transport Layer

- Responsible for process-to-process (end-to-end) delivery of the entire message, in units called **segments**.
- Provides **segmentation, flow control, and error control**.
- Two main protocols: **TCP** (connection-oriented, reliable) and **UDP** (connectionless, fast).

Function	Description
Segmentation & Reassembly	Divides message into segments at sender; reassembles them at receiver
Service-point/Port Addressing	Uses port numbers to deliver data to the correct process/application
Connection Control	Establishes, maintains, and terminates connections (connection-oriented mode)
Flow Control	End-to-end flow control between sender and receiver processes
Error Control	End-to-end error checking, acknowledgements, and retransmission

Devices/Protocols: Gateways; TCP, UDP, SCTP.

Unit of Data: Segment (TCP) / Datagram (UDP)

Layer 5: Session Layer

- Establishes, manages, and terminates **sessions** between two communicating applications.
- Provides **dialog control** (half-duplex/full-duplex) and synchronization.

Function	Description
Session Establishment	Sets up, coordinates, and terminates conversations (sessions) between applications
Dialog Control	Allows communication in half-duplex or full-duplex mode
Synchronization	Adds checkpoints to a stream of data so that, in case of failure, transmission can resume from the last checkpoint instead of starting over

Devices/Protocols: NetBIOS, PPTP, RPC, SIP.

Unit of Data: Data

Layer 6: Presentation Layer

- Concerned with the syntax and semantics of data exchanged between systems.
- Handles **translation, encryption/decryption, and compression** of data.

Function	Description
Translation	Converts data between different formats/encodings used by sender and receiver systems (e.g., ASCII to EBCDIC)
Encryption/Decryption	Ensures privacy by transforming data before transmission and restoring it at the receiver
Compression	Reduces the number of bits to be transmitted, improving efficiency

Devices/Protocols: SSL/TLS, JPEG, MPEG, GIF.

Unit of Data: Data

Layer 7: Application Layer

- Provides network services directly to end-user applications.
- Examples: HTTP, FTP, SMTP, DNS, Telnet.

Function	Description
Network Virtual Terminal	Allows a user to log on to a remote host (e.g., Telnet)
File Transfer / Access / Management	Allows access, retrieval, and management of files on a remote computer
Mail Services	Provides the basis for email forwarding and storage
Directory Services	Provides distributed database sources for global information about objects/services (e.g., DNS)

Devices/Protocols: HTTP, HTTPS, FTP, SMTP, POP3, DNS, Telnet.

Unit of Data: Data

3.4 Networking Devices at Each OSI Layer

Layer	Associated Devices
Application / Presentation / Session	Gateways, end-user software (browsers, mail clients)
Transport	Gateways, firewalls (stateful)
Network	Routers, Layer-3 switches
Data Link	Switches, Bridges, NIC
Physical	Hubs, Repeaters, Cables, Connectors

3.5 Merits and Demerits of OSI Model

Merits	Demerits
Clear, modular, layer-by-layer division of network functions	Purely theoretical; not directly implemented in real networks
Helps in standardization and interoperability between vendors	More complex than the simpler 4-layer TCP/IP model
Simplifies troubleshooting by isolating faults to a specific layer	Some layers (e.g., Session, Presentation) are thin/merged in practice
Useful as a teaching and reference framework	Slower to adapt compared to protocol suites developed for the real Internet

3.6 Data Encapsulation Process

As data moves down the layers at the sender's side, each layer adds its own header (and sometimes trailer) — this process is called **encapsulation**. At the receiver's side, the reverse process, **decapsulation**, strips off the headers layer by layer.

Step-by-Step Encapsulation Example

1. **Application layer** creates the user message (e.g., an email) → forms **Data**.
2. **Presentation layer** formats/encrypts/compresses the data (still **Data**, no new PDU name).

3. **Session layer** adds session/dialog-control information to manage the conversation (still **Data**).
4. **Transport layer** adds a header (with port numbers) → forms a **Segment**.
5. **Network layer** adds an IP header (with source/destination IP) → forms a **Packet**.
6. **Data Link layer** adds a header and trailer (with MAC addresses and CRC) → forms a **Frame**.
7. **Physical layer** converts the frame into a stream of **Bits** for transmission over the medium.

At the receiving end, each layer removes its corresponding header/trailer in reverse order (decapsulation) until the original data reaches the destination application.

Note: Application, Presentation, and Session layers all work on the same PDU called **Data** — no new header name is introduced until the Transport layer, which is why some textbooks group them together as the "upper layers."

What is PDU?

PDU stands for **Protocol Data Unit** — the name given to a unit of data at a specific layer of a network model. As data passes down the layers, each layer adds its own header/trailer, and the data unit is referred to by a different name (PDU) at each layer, as shown below.

Layer	Data Unit (PDU)
Application/Presentation/Session	Data
Transport	Segment
Network	Packet
Data Link	Frame
Physical	Bits

3.7 Summary Table of All Layers

Layer No.	Layer Name	Key Function	PDU	Example Protocols
7	Application	Network services to end-user apps	Data	HTTP, FTP, SMTP, DNS
6	Presentation	Translation, encryption, compression	Data	SSL/TLS, JPEG, MPEG
5	Session	Session establishment, dialog control	Data	NetBIOS, PPTP
4	Transport	End-to-end delivery, error/flow control	Segment	TCP, UDP
3	Network	Logical addressing, routing	Packet	IP, ICMP
2	Data Link	Framing, MAC addressing, error detection	Frame	Ethernet, PPP
1	Physical	Transmission of raw bits	Bits	Cables, hubs, NIC

Memory Trick (Mnemonic)

"All People Seem To Need Data Processing" — Application, Presentation, Session, Transport, Network, Data Link, Physical (Layer 7 down to Layer 1).

Chapter Summary

The OSI model has 7 layers — Physical, Data Link, Network, Transport, Session, Presentation, and Application — each performing a specific function. Encapsulation adds headers as data passes down the layers.

Chapter 4: Network Topologies

4.1 Introduction

Network Topology refers to the arrangement/layout of nodes (computers, devices) and connecting links in a network. It can be physical (actual layout of cables/devices) or logical (the way data flows).

Type	Description
Physical Topology	The actual physical layout/wiring of cables, devices, and nodes
Logical Topology	The way data actually flows within the network, regardless of physical layout

Factors Affecting Choice of Topology

- **Cost** — cabling and installation expenses.
- **Scalability** — ease of adding new nodes in future.
- **Reliability** — impact of a single link/node failure on the whole network.
- **Ease of Installation & Maintenance** — simplicity of setting up and troubleshooting.
- **Bandwidth/Performance Requirement** — amount of traffic the network needs to support.

4.2 Bus Topology

All devices are connected to a single central cable called the **backbone** or **bus**. Data sent by any device travels along the bus in both directions until it reaches the destination; terminators at each end absorb the signal to prevent reflection.

- Uses a single shared communication medium for all devices.
- Requires **terminators** at both ends of the backbone to stop signal bounce-back.
- Suitable for small, temporary networks with few nodes.

Advantages	Disadvantages
Easy and cheap to install (least amount of cabling)	Entire network goes down if the backbone cable breaks
Works well for small networks	Difficult to detect faults and troubleshoot
Requires fewer cables compared to star/mesh	Performance degrades as more devices/traffic are added
Easy to extend by adding nodes along the bus	Limited cable length and number of nodes supported

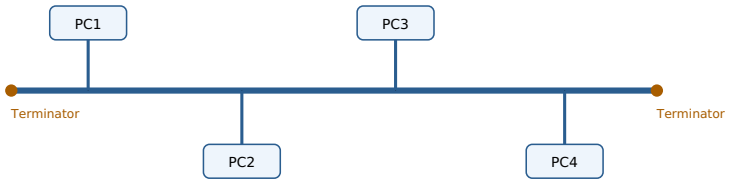


Fig. 4.1 — Bus Topology

4.3 Star Topology

All devices are connected to a central device (hub/switch); all data passes through the centre. It is the most commonly used physical topology in modern LANs.

- Central device (hub/switch) manages and directs all communication.
- If the central device supports it, devices can communicate at full speed without collisions (when using a switch).
- Easiest topology to design, install, and reconfigure.

Advantages	Disadvantages
Easy to install, manage, and troubleshoot	Entire network fails if the central hub/switch fails (single point of failure)
Failure of one node/cable does not affect other nodes	Requires more cabling than bus topology
Easy to add or remove devices	Cost depends on the capability of the central device
Centralized management and monitoring	Network size limited by the number of ports on the central device

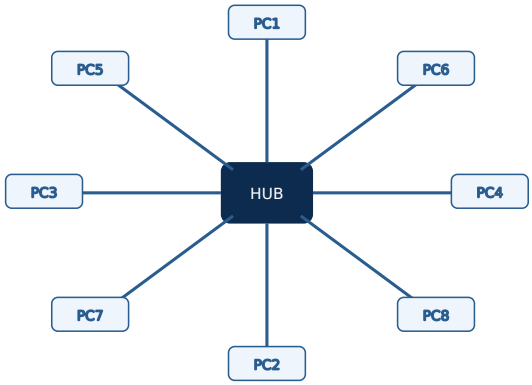


Fig. 4.2 — Star Topology

4.4 Ring Topology

Each device is connected to exactly two neighbouring devices, forming a closed loop/ring. Data travels in one direction (or both, in dual ring), passing through each node until it reaches its destination.

- Each node acts as a **repeater**, regenerating the signal and forwarding it to the next node.
- **Token passing** is often used to control access to the ring (e.g., Token Ring, FDDI).
- **Dual-ring** — a configuration consisting of two concentric rings carrying data in opposite directions — variants provide a backup path if one ring fails.

What is Token Passing?

Token Passing is a media access control method in which a special control frame called a **token** circulates around the network in a fixed order. Only the device holding the token is permitted to transmit data; after transmitting (or if it has nothing to send), it passes the token to the next device. This ensures fair, organized, and collision-free access to the shared network medium.

Advantages	Disadvantages
Equal access opportunity for all nodes	Failure of a single node/cable can disrupt the entire ring (unless dual ring)
Easier fault isolation than bus topology	Adding/removing a node requires temporarily breaking the ring
No central device required (no single point of failure at the centre)	Overall network speed limited by the slowest link/node
Data transmission is orderly and collision-free (with token passing)	Troubleshooting can be difficult in large rings

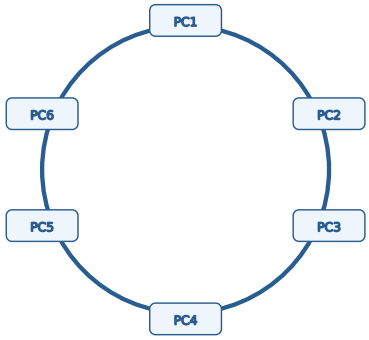


Fig. 4.3 — Ring Topology

4.5 Mesh Topology

Every device is connected to every other device through dedicated point-to-point links. Provides maximum redundancy and is mainly used where reliability is critical (e.g., backbone of WANs, military/defense networks).

Type	Description
Full Mesh	Every node is connected directly to every other node — n(n−1)/2 links required for n nodes
Partial Mesh	Only some nodes are interconnected directly; others connect indirectly through intermediate nodes

Advantages	Disadvantages
Highly reliable — failure of one link does not affect the rest of the network	Very expensive due to large amount of cabling and ports required
Provides high data security and privacy on dedicated links	Complex to install, configure, and maintain
Easy fault identification and isolation	Not practical for large networks (cabling grows rapidly: n(n−1)/2)
No traffic congestion problem since each link is dedicated	Requires significant physical space for cabling and ports

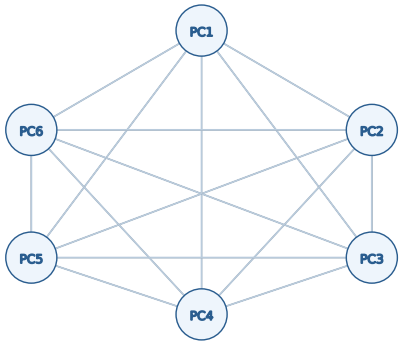


Fig. 4.4 — Mesh Topology (every node interconnected)

4.6 Tree Topology

A hierarchical structure combining characteristics of bus and star topology — devices are organized in a parent-child (hierarchical) layout, often used in WAN/enterprise networks and to interconnect multiple star networks.

- Has a **root node** at the top, with branches extending to secondary hubs/switches and then to end devices.
- Suitable for organizations with a hierarchical structure (e.g., head office → branch offices → departments).

Advantages	Disadvantages
Scalable — easy to add new branches/nodes	If the root/backbone fails, the entire connected sub-network fails
Supports hierarchical management and easy fault isolation per branch	Complex wiring and configuration as network grows
Good for organizing larger networks into manageable segments	Maintenance cost increases with the number of hierarchy levels

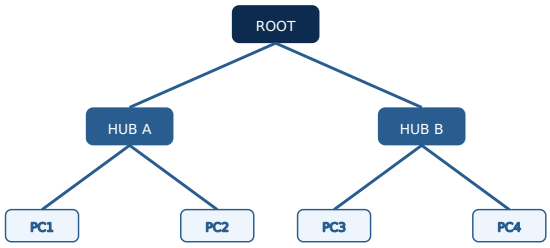


Fig. 4.5 — Tree Topology

4.7 Hybrid Topology

A combination of two or more different topologies (e.g., Star-Bus, Star-Ring) used to leverage the advantages of each while minimizing their disadvantages. Common in large, real-world organizational networks where different departments may use different topologies internally, connected together.

Advantages	Disadvantages
Flexible — combines the benefits of the topologies used	Complex design and configuration
Scalable for large, growing organizations	Costlier to implement and maintain than a single topology
Faults can often be isolated to a specific segment	Requires skilled personnel for design and troubleshooting

4.8 Point-to-Point Topology

The simplest topology, where two devices are connected directly by a single dedicated link, with no other devices sharing the connection.

Advantages	Disadvantages
Simple to set up; full bandwidth available between the two devices	Not scalable — connects only two devices
High security since the link is dedicated	Not suitable for networks with many devices

4.9 Comparative Table of Topologies

Topology	Advantages	Disadvantages
Bus	Easy to install; low cable cost	Entire network fails if backbone cable breaks; difficult to troubleshoot
Star	Easy to install/manage; failure of one node doesn't affect others	Entire network fails if hub/switch fails; more cabling required
Ring	Equal access for all nodes; simple data flow	Failure of one node/link can disrupt entire ring (unless dual ring)
Mesh	Highly reliable; robust against link failure; high security	Very expensive; complex cabling; difficult to install/maintain

Tree	Scalable; supports hierarchical management	Heavily dependent on root/backbone; complex wiring
Hybrid	Flexible; combines benefits of multiple topologies	Complex design; costly to implement
Point-to-Point	Simple; full dedicated bandwidth; highly secure	Not scalable beyond two devices

Chapter Summary

Network topology describes the physical/logical layout of a network. Common topologies are Bus, Star, Ring, Mesh, Tree, and Hybrid — each with trade-offs in cost, reliability, and ease of maintenance.

Chapter 5: Network Protocols

5.1 Definition and Importance

A **protocol** is a set of rules that governs how data is formatted, transmitted, and received between communicating devices on a network. Protocols ensure that devices from different manufacturers can communicate effectively.

5.2 Characteristics / Key Elements of Protocols

Every protocol is defined by three key elements:

Element	Description
Syntax	The structure/format of data — order of bits/fields
Semantics	The meaning of each section of bits; how a pattern is interpreted and what action is to be taken
Timing	When data should be sent and how fast it can be sent (speed/synchronization)

5.3 Types of Network Protocols

Protocol	Function
TCP (Transmission Control Protocol)	Reliable, connection-oriented delivery of data
IP (Internet Protocol)	Logical addressing and routing of packets
UDP (User Datagram Protocol)	Connectionless, fast but unreliable delivery
HTTP/HTTPS	Transfer of web pages (Hypertext Transfer Protocol)
FTP (File Transfer Protocol)	Transfer of files between systems
SMTP (Simple Mail Transfer Protocol)	Sending email messages
DNS (Domain Name System)	Translates domain names to IP addresses
Ethernet	LAN-level data link/physical layer protocol

5.4 Role of Protocols in Communication

- Ensure correct formatting and interpretation of transmitted data.
- Manage error detection and correction.
- Handle flow control to prevent data overload.
- Establish, maintain, and terminate connections between devices.
- Provide addressing schemes to identify devices uniquely.

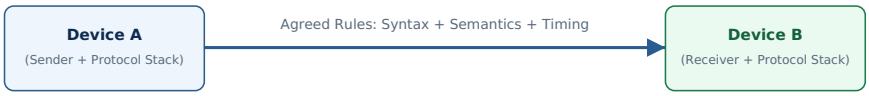


Fig. 5.1 — Protocol Communication Process

Chapter Summary

A protocol is a rule-set governing communication, defined by syntax, semantics, and timing. Common protocols include TCP, IP, UDP, HTTP, FTP, SMTP, and DNS.

Chapter 6: Ethernet

6.1 Introduction and History

Ethernet is the most widely used LAN technology, originally developed by **Xerox PARC** in 1973 (Robert Metcalfe), and later standardized as **IEEE 802.3**. It operates at the Physical and Data Link layers of the OSI model.

6.2 Features of Ethernet

- Uses **CSMA/CD** (Carrier Sense Multiple Access with Collision Detection) as its access method.
- Supports bus and star topologies (modern Ethernet mostly uses star with switches).
- Speeds range from 10 Mbps (legacy) to 10/40/100 Gbps (modern Ethernet).
- Uses **MAC addresses** for identifying devices on the network.
- Standardized under **IEEE 802.3**.

6.3 Working Principle

In traditional (shared-medium) Ethernet, all devices share the same communication channel. Before transmitting, a device "listens" to check if the channel is free; this principle is implemented through CSMA/CD.

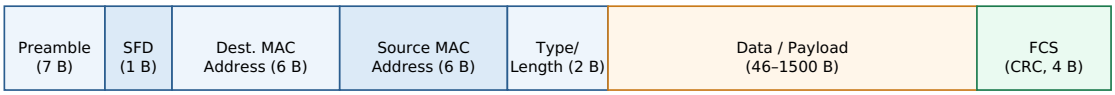
6.4 CSMA/CD Concept

Carrier Sense Multiple Access with Collision Detection (CSMA/CD) works as follows:

1. **Carrier Sense** – the device checks whether the channel is idle (free) before transmitting.
2. **Multiple Access** – multiple devices share the same transmission medium.
3. **Collision Detection** – if two devices transmit simultaneously, a collision occurs; both devices detect it, stop transmitting, and wait a random time (back-off) before retrying.

> **Exam Point**
> CSMA/CD is used in traditional (half-duplex) Ethernet. Modern full-duplex switched Ethernet does not require CSMA/CD since there are no collisions in dedicated point-to-point links.

6.5 Ethernet Frame Format



SFD = Start Frame Delimiter | FCS = Frame Check Sequence

Fig. 6.1 — Ethernet Frame Structure

Field	Size	Purpose
Preamble	7 bytes	Syncs sender/receiver timing so bits are read correctly
Start Frame Delimiter (SFD)	1 byte	Marks where the actual frame begins
Destination MAC Address	6 bytes	Says whom the frame is for
Source MAC Address	6 bytes	Says who sent the frame
Type/Length	2 bytes	Tells what kind of data is inside
Data/Payload	46–1500 bytes	The actual message being carried
FCS (CRC)	4 bytes	Checks if the frame got damaged in transit

6.6 Advantages and Limitations of Ethernet

Advantages	Limitations
Inexpensive and easy to install	Performance degrades with collisions in shared mode
Reliable and widely supported	Limited cable length without repeaters
Scalable (10 Mbps to 100+ Gbps)	Shared bus version suffers from congestion
Simple frame structure and error checking	Less suitable for very long distances (used mainly in LAN)

6.7 Applications of Ethernet

- Office and campus LANs.
- Data centre networking.

- Home networking (wired connections).
- Backbone connectivity for larger networks.

Chapter Summary

Ethernet (IEEE 802.3) is the dominant LAN technology, using CSMA/CD for shared-medium access and a standard frame format with MAC addressing and CRC-based error checking.

Chapter 7: ARCNET

7.1 Introduction

ARCNET (Attached Resource Computer NETwork) is one of the earliest LAN technologies, introduced by Datapoint Corporation in 1977. It uses a **token-passing** access method over a star or bus topology.

7.2 Features and Characteristics

- Uses **token passing** for media access control (deterministic, collision-free).
- Can be wired in a star, bus, or distributed star topology using coaxial, twisted-pair, or fibre cable.
- Original data rate: 2.5 Mbps (later versions, ARCNET Plus, support up to 20 Mbps).
- Standardized as **ANSI/ATA 878.1**.

7.3 Working Principle / Token Passing Mechanism

In token passing, a special control frame called a **token** circulates among the nodes in a predefined (logical) order. Only the node holding the token is permitted to transmit data; once finished, it passes the token to the next node in sequence. This guarantees fair, collision-free access to the network.

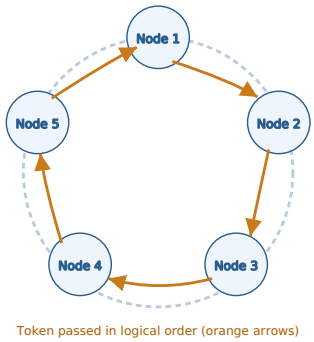


Fig. 7.1 — ARCNET Network Structure (Token Passing)

7.4 Advantages and Disadvantages

Advantages	Disadvantages
Collision-free, deterministic access	Lower data rate compared to modern Ethernet
Predictable performance under heavy load	Largely obsolete; replaced by Ethernet
Simple, reliable, low-cost wiring	Limited scalability and vendor support
Good for real-time/industrial control applications	Slower token rotation as nodes increase

7.5 Applications

- Industrial automation and control systems (valued for deterministic timing).
- Early small-business and educational LANs (historical use).
- Embedded/real-time systems requiring predictable network access.

Chapter Summary

ARCNET is an early LAN technology using deterministic token-passing access, offering collision-free but comparatively slower communication than Ethernet.

Chapter 8: TCP/IP Model

8.1 Introduction

The **TCP/IP model (Internet Protocol Suite)** was developed by the **US Department of Defense (DoD)** in the 1970s and is the foundation of the modern Internet. Unlike the 7-layer OSI model, TCP/IP is organized into **four layers**.

8.2 TCP/IP Architecture

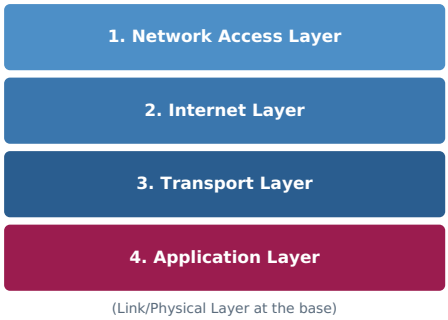


Fig. 8.1 — TCP/IP Model (4 Layers)

8.3 Four Layers of TCP/IP

1. Network Access Layer (Link Layer)

- Combines functions of OSI's Physical and Data Link layers.
- Responsible for placing packets onto the physical network medium and handling hardware addressing (MAC).
- Examples: Ethernet, Wi-Fi (IEEE 802.11), ARCNET, PPP.

2. Internet Layer

- Corresponds to OSI's Network layer.
- Responsible for logical addressing (IP addresses) and routing of packets across networks.
- Key protocols: **IP**, **ICMP**, **ARP**.

3. Transport Layer

- Same role as OSI's Transport layer — provides end-to-end communication.
- Key protocols: **TCP** (reliable, connection-oriented) and **UDP** (unreliable, connectionless).

4. Application Layer

- Combines OSI's Session, Presentation, and Application layers.
- Provides protocols used directly by user applications.
- Examples: HTTP, FTP, SMTP, DNS, Telnet, SNMP.

8.4 TCP vs IP

Aspect	TCP	IP
Full Form	Transmission Control Protocol	Internet Protocol
Layer	Transport Layer	Internet/Network Layer
Function	Ensures reliable, ordered, error-checked delivery of data	Provides logical addressing and routing of packets
Connection Type	Connection-oriented	Connectionless
Reliability	Reliable (acknowledgements, retransmission)	Best-effort, unreliable on its own

8.5 OSI vs TCP/IP Comparison

Basis	OSI Model	TCP/IP Model
Number of Layers	7 Layers	4 Layers
Developed By	ISO	US Department of Defense (DARPA)
Approach	Theoretical / Conceptual reference model	Practical / Implementation-based model
Layer Independence	Each layer strictly independent	Layers somewhat interdependent
Usage	Used as a reference/teaching model	Used as the actual protocol suite of the Internet
Reliability Focus	Reliability handled at multiple layers	Reliability mainly handled at Transport layer (TCP)

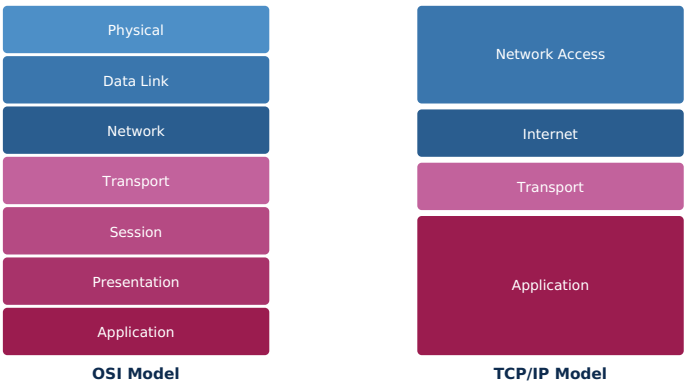


Fig. 8.2 — OSI Model vs TCP/IP Model (Layer Mapping)

Chapter Summary

TCP/IP is a 4-layer, practical protocol suite (Network Access, Internet, Transport, Application) that forms the basis of the Internet, in contrast to the 7-layer theoretical OSI model.

Chapter 9: Quick Revision Section

9.1 Important Definitions (One-Liners)

Term	One-Line Definition
Network	Interconnected devices that share data and resources
LAN	High-speed network covering a small geographical area
OSI Model	7-layer conceptual model for standardizing network communication
Topology	Physical/logical arrangement of nodes in a network
Protocol	Rules governing format, meaning, and timing of communication
Ethernet	IEEE 802.3 LAN standard using CSMA/CD and MAC addressing
ARCNET	Token-passing LAN technology, collision-free access
TCP/IP	4-layer protocol suite forming the basis of the Internet

9.2 Key Features at a Glance

OSI Layers (Top → Bottom)

7. Application
6. Presentation
5. Session
4. Transport
3. Network
2. Data Link
1. Physical

TCP/IP Layers (Top → Bottom)

4. Application
3. Transport
2. Internet
1. Network Access

9.3 Important Differences

Comparison	Point of Difference
OSI vs TCP/IP	7 layers (theoretical) vs 4 layers (practical, Internet standard)
TCP vs UDP	Connection-oriented & reliable vs Connectionless & fast
Ethernet vs ARCNET	CSMA/CD (contention-based) vs Token passing (deterministic)
LAN vs WAN	Small area, high speed, low cost vs Large area, lower speed, higher cost
Bus vs Star Topology	Single shared cable vs central hub/switch connecting all nodes

9.4 One-Page Revision Notes

Rapid Revision

- 5 components of data communication: Message, Sender, Receiver, Medium, Protocol.
- OSI model mnemonic: "All People Seem To Need Data Processing" (Layer 7 → 1).
- PDU names: Data (App/Pres/Session) → Segment (Transport) → Packet (Network) → Frame (Data Link) → Bits (Physical).
- Topologies: Bus, Star, Ring, Mesh, Tree, Hybrid.
- Ethernet uses CSMA/CD; standardized as IEEE 802.3.
- ARCNET uses Token Passing; collision-free, deterministic.
- TCP/IP has 4 layers: Network Access, Internet, Transport, Application.
- TCP = reliable & connection-oriented; IP = addressing & routing.

9.5 Frequently Asked Exam Questions

Q1. Who developed the OSI model and when?
A. The OSI model was developed by ISO (International Organization for Standardization) in 1984.

Q2. Name the seven layers of the OSI model in order.
A. Physical, Data Link, Network, Transport, Session, Presentation, Application.

Q3. Which access method does Ethernet use?
A. CSMA/CD (Carrier Sense Multiple Access with Collision Detection).

Q4. Which access method does ARCNET use?
A. Token Passing.

Q5. How many layers does the TCP/IP model have? Name them.

A. Four layers — Network Access, Internet, Transport, Application.

Q6. What is the difference between TCP and UDP?

A. TCP is connection-oriented and reliable; UDP is connectionless and faster but unreliable.

Q7. What are the three key elements of a protocol?

A. Syntax, Semantics, and Timing.

Q8. Which topology connects every device to every other device directly?

A. Mesh Topology.

Q9. What is the standard name of Ethernet under IEEE?

A. IEEE 802.3.

Q10. Define LAN.

A. A Local Area Network is a high-speed network confined to a small geographical area such as a building or campus, typically privately owned.

End of Notes — Basic Concept of Networking and Data Communications. Best of luck for your examination!